# VOXBLINK: X-LARGE SPEAKER VERIFICATION DATASET ON CAMERA

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## ABSTRACT

In this paper, we contribute a novel and extensive dataset for speaker verification, which contains noisy 38k identities/1.45M utterances (VoxBlink) and relatively cleaned 18k identities/1.02M (VoxBlink-Clean) utterances for training. Firstly, we accumulate a 60K+ users' list with their avatars and download their short videos on YouTube. We then established an automatic and scalable pipeline to extract relevant speech and video segments from these videos. To our knowledge, the VoxBlink dataset is one of the largest speaker recognition datasets available. Secondly, we conduct a series of experiments based on different backbones trained on a mix of the VoxCeleb2 and the VoxBlink-Clean. Our findings highlight a notable performance improvement, ranging from 15% to 30%, across different backbone architectures upon integrating our dataset for training. The dataset will be made publicly available shortly.

***Index Terms—*** Speaker Verification, Dataset, Large-scale, Multimodel.

## 1. INTRODUCTION

Speaker verification in the wild has achieved remarkable success due to the boom of CNNs. The key drivers behind the success of automatic speaker verification (ASV) consists of the evolution of backbone architecture[1, 2, 3, 4], the introduction of diverse loss functions[5, 6, 7], and the availability of large-scale corpora[8, 9]. Even though growing efforts have been devoted to refining networks and training strategies, the academic community still faces constraints by the limited number and diversity of available training sets.

We can make a comparison with facial datasets, millions of images establish a robust foundation for facial recognition applications. Regrettably, in the field of speaker recognition, the availability of publicly accessible datasets with both mountainous utterances and speakers in the wild remains noticeably limited. As is shown in Fig.1, many contributions have been made to enrich ASV datasets[8, 9, 10, 11, 12, 13, 14, 13]. Some datasets[11, 12] mainly comprise multi-channel speeches, which reduces the diversity of training data. While others[10, 13, 12] fall short of their limited styles, languages and scales. Among these endeavors, the VoxCeleb [8, 9] stands out as the most successful as it contains over one million utterances from thousands of speakers. Nonetheless, its exclusive reliance on celebrity data limits its scalability. Furthermore, when compared to facial recognition datasets, the VoxCeleb's scale is considerably smaller.

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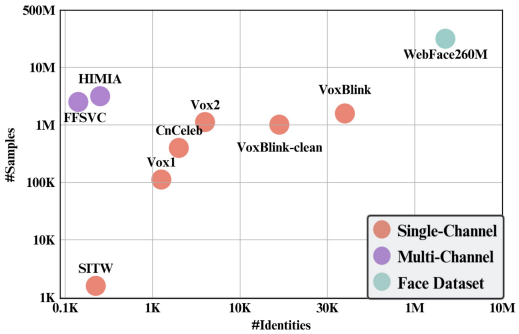


**Fig. 1**. Comparisons of # identities and # samples for our VoxBlink data and public ASV training set (also a face dataset as a contrast). The x and y-axis have **non-uniform** scales.

To address this gap, we introduce the VoxBlink, an x-large scale dataset for speaker verification. Unlike the VoxCeleb, all the data in the VoxBlink is captured automatically from users who upload *short* videos on YouTube, which contain over 1.4M utterances from over 30K speakers. Moreover, By utilizing refined inference models and relying on more scalable data sources, our pipeline is more robust and promising than the VoxCeleb's. The speech segments share cross-age, cross-lingual, cross-channel and cross-device attributes as the dataset is formulated "in the wild". This dataset is also audio-visual, potentially valuable for other applications like speech separation[15, 16], multi-modal verification[17] and diarization[18, 19], etc.

In the second part of the paper, we incorporate the VoxBlink and train various models with different backbones. When introducing VoxBlink-clean, all models exhibit a performance boost ranging from 15% to 30%. Our primary contributions can be summarized as:

(i) We propose a more scalable and robust pipeline to collecting speaker recognition data. (ii) We collect a novel x-large speaker recognition dataset VoxBlink and its purified version, VoxBlink-clean. (iii) We achieve a considerable performance improvement under different backbones by integrating the VoxBlink-clean.

## 2. THE VOXBLINK DATASET

### 2.1. Data Description

In the internet era, everyone serves as a data provider. The VoxBlink contains contains 1,455,237 uttearnces from 38,067 users in YouTube. All segments are extracted from short videos uploaded

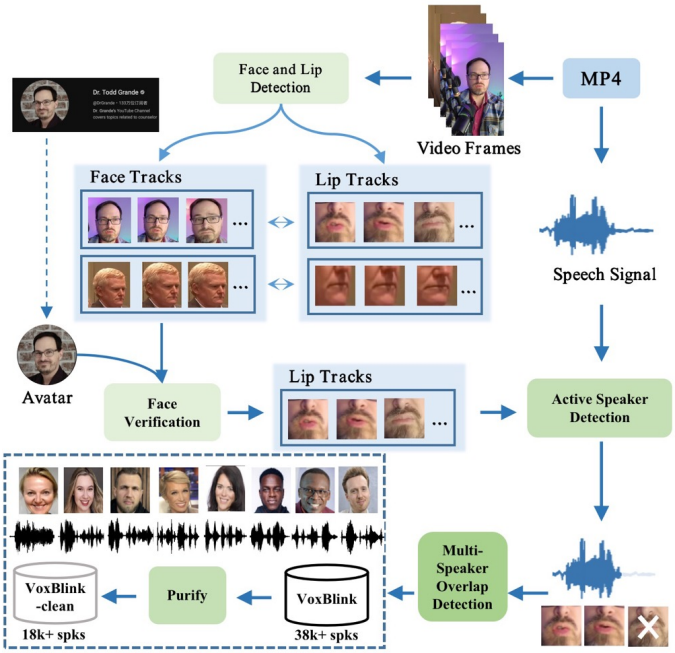


**Fig. 2**. The automatic pipeline (step II∼V) for the VoxBlink

by ordinary users on YouTube, encompassing a wide variety of contexts including podcasts, vlogs, speeches, live streaming highlights, etc. Indoor reverberation, non-verbal sounds, background music and other acoustic conditions have increased the complexity and diversity of the data. Most of the segments are recorded on mobile devices, with recording environments spanning indoors, outdoors, and a variety of complex scenarios. Other general information can be referred to Table.1 and Fig.3 shows a visualization of the statistics.

### 2.2. PipeLine

As shown in Fig.2, we use an automatic multi-modal pipeline to curate VoxBlink from Youtube. The main innovation is that since the abundance of users' avatars containing their faces, we can perform frame-by-frame face verification between their avatars and the videos they upload. Additionally, with the help of other tools, we can extract speech/video segments specifically pertaining to the target user. For clarity, we summarize this process as follows:

**Step I: Candidate list and video downloading.** We start by preparing a list of over 2,000 commonly used names for YouTube searches. Having observed that users are more likely to appear in short videos, we opt to retain only those users who have the avatar with a single face and who have uploaded short videos. To enhance diversity, we also incorporated several professions and themes into our search keywords. Over one million videos from 61,038 users with their avatars are downloaded after duplicate removal in the Youtube retrieval.

**Step II: Face and lip tracking.** The Retina Face[20] is adopted as face detector as well as lip movement detector to output video and lip tracks. By setting a threshold for the minimum Intersection over Union (IOU) value between two consecutive detections, we ensure that each track contains the face of only one individual.

**Step III: Face verification.** We utilize ResNet-IRSE50[6] to extract

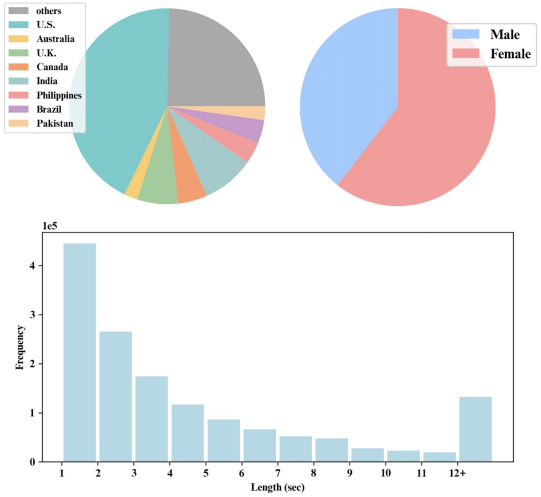


**Fig. 3**. **Top:** the distribution of gender (*left*) and geographic location (*left*) of speakers. **Bottom**: The distribution of utterance lengths in the dataset

speaker embeddings frame by frame along the track. Meanwhile, the face embedding of the avatar has been extracted as template embedding after flip augmentation, which promotes the robustness. Then cosine scores are calculated along the track and the track-level average score is calculated to discard non-target tracks.

**Step IV: Active Speaker Detection.** This process is employed to acquire genuine speech segments of the speaker from the tracks. An Seq2Seq audio-visual active speaker detection model[21] aids to locate active speech by lip motions and audio in the track. This method not only helps to discard the silent or voice-over pieces, but also reject out-of-sync fragments.

**Step V: Overlap Detection.** The goal of this step is to exclude overlap speech segments, which would interfere the training performance. Another conformer-based toolkit, which has been trained on our self-synthesized dataset, is adopted here to further enhance the quality of speech segments. Additionally, we abandon these utterances with an extremely short duration (less than 1 seconds).

**Step VI: Meta info collection.** Due to platform constraints, only the geographical locations of approximately 21,000 speakers are recorded, with participants spanning 130 different regions. Given the challenges in obtaining gender labels, we binary classified the speakers' genders using audio-visual data. As is shown in Fig.3, about 60.5% speakers are female. We also collect other meta information about the video including release time, category, and tags of each video for other potential application.

Additionally, we only preserve those speakers with 5+ utterances to maintain the with-in speaker diversity. The audio-visual data obtained through the pipeline constitutes VoxBlink, whose name is inspired by the characteristics of short videos. The VoxBlink is only designed to be a training-only set, thus the thresholds in the pipeline are set to be less strict manually.

### 2.3. Purify

Since the threshold has been manually set, there may be some noise data present unfortunately. In addition, during our manual listening, we discovered some recordings where users are merely lip-syncing

**Fig. 4**. The wordcloud that highlights the thematic diversity of the VoxBlink, which is generated by videos' tags.

**Table 1**. *Dataset statistics for the VoxBlink.*

| Dataset | VoxBlink | VoxBlink-clean |
|---|---|---|
| # of SPKs | 38,067 | 18,381 |
| # of videos | 372,091 | 241,172 |
| # of hours | 2,135 | 1,670 |
| # of utterances | 1,455,237 | 1,028,106 |
| Avg # of videos per SPK | 9.77 | 13.12 |
| Avg # of utterances per SPK | 38.23 | 55.93 |
| Avg # of duration per utterance (s) | 5.28 | 4.87 |

rather uttering, indicating a need for further purifying of our data. We purify the VoxBlink dataset based on the following 3 scores:

- The avarage score of within-speaker embedding cosine similarities derived by ResNet34[2].
- The avarage score of within-speaker face embedding cosine similarities derived by [6].
- The average music-speech discrimination score of a speaker by [22].

The purified VoxBlink is named VoxBlink-clean, where we find few noise labels during the audition.

## 3. SPEAKER VERIFICATION SETTINGS

In this section, we describe the experimental settings of several speaker verification systems. Evaluation metrics and implemental details are detailly introduced to follow. We also suggest a Mix-Finetune(Mix-FT) training mode to accelerate the training process.

### 3.1. Evaluation metrics

Since the VoxBlink serves as a training-only dataset, we follow the VoxCeleb1-based and VoxSRC23-dev protocols. Two performance metrics are taken into account: (i)Equal Error Rate (EER) represents the rate at which both acceptance and rejection errors are equal. (ii) minDCF (Minimum Detection Cost Function) shows the minimized detection cost considering different decision costs and error rates. Under our testing scenario, $p_{target}$ is set 0.05.

### 3.2. Implement details

We utilize VoxCeleb2 and VoxBlink-clean to achieve the optimal results. On-the-fly data augmentation[23] to add additive background noise or convolutional reverberation noise for the time-domain waveform. Also, we apply speaker augmentation with speed perturbation[24]. We speed up or down each utterance by a factor of 0.9 or 1.1, yielding shifted pitch utterances that are considered from new speakers. Finally the training data containing 6,360,345 utterances (3,084,318 from the VoxBlink and 3,276,027 from the VoxCeleb2) from 73,125 speakers (55,143 from the VoxBlink and 17,982 from the VoxCeleb2). The acoustic features are 80-dimensional log Mel-filterbank energies with a frame length of 25ms and hop size of 10ms. The input frame length is fixed at 200 frames and cosine similarity scores are calculated in the evaluation phase. As for the back-end head, we utilize the AAM-Softmax[6] (m=0.2, s=32) to classify different speakers.

Inspired by [25], we observed that starting the training with a blend of both datasets from the onset, as compared to initially training with VoxCeleb2 for a period followed by incorporating VoxBlink-clean, yields closed performances. Therefore we carry most of our experiments based on three stages:

In the first stage, we perform a linear warm-up learning rate schedule at the first 5 epochs to the initial learning rate at 0.1, which aims to prevent model vibration and speed model training. In the second stage, the SGD optimizer is employed to update the model parameters and the StepLR scheduler with 0.1 initial LR drop to 1e-3. As the first two stages only use VoxCeleb2, we introduce VoxBlink-clean in the last two phases. The training process resumes at 1e-3 lr and gradually drops till convergence.

### 3.3. Model Settings

**ResNet-Based.** Systems based on ResNet consistently excel in classification tasks. We primarily adopted two structures based on ResNet. The first We utilize ResNet34 [26] followed by a temporal statistic pooling layer. The widths (channel number) of the residual blocks are{ 64, 128, 256, 512}. Another model is ResNet100 with frequency-wise Squeeze-Excitation (fwSE). [27] blocks, which has a relative larger scale.

**SimAM-ResNet.** ResNet with Simple attention module (SimAM) has presented strong competitiveness in ASV[28]. The SimAM module is a plug-and-play module without extra modal parameters. Here we adopt ResNet34 and ResNet100 with SimAM followed by attentive statistics pooling(ASP).

**ECAPA-TDNN.** ECAPA-TDNN[1] is currently the most popular and state-ofthe-art (SOTA) TDNN-series model for speaker verification. We use the 1024-channel ECAPA-TDNN for testing here.

**Table 2**. *Experimental results of different backbones with/without the VoxBlink. It should be noted that we do not use any post-processing method and all the benchmarks are based on the cosine scores between trails.*

| ID | Model | VoxBlink-clean | Vox-O | | Vox-E | | Vox-H | | VoxSRC23-dev | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | EER | $mDCF_{0.01}$ | EER | $mDCF_{0.01}$ | EER | $mDCF_{0.01}$ | EER | $mDCF_{0.05}$ |
| **M1** | ResNet34 | × | - | - | - | - | - | - | - | - |
| | | ✓ | - | - | - | - | - | - | - | - |
| **M2** | FwSE-ResNet100 | × | 0.580 | 0.057 | 0.775 | 0.083 | 1.438 | 0.141 | 3.923 | 0.228 |
| | | ✓ | 0.452 | 0.038 | 0.709 | 0.079 | 1.277 | 0.128 | 3.795 | 0.209 |
| **M3** | SimAM-ResNet34 | × | 0.767 | 0.066 | 0.894 | 0.100 | 1.626 | 0.162 | 4.153 | 0.227 |
| | | ✓ | 0.638 | 0.070 | 0.833 | 0.096 | 1.539 | 0.156 | 4.247 | 0.231 |
| **M4** | SimAM-ResNet100 | × | 0.596 | 0.053 | 0.761 | 0.083 | 1.391 | 0.132 | 3.727 | 0.207 |
| | | ✓ | 0.441 | 0.044 | 0.681 | 0.075 | 1.268 | 0.125 | 3.549 | 0.197 |
| **M5** | ECAPA-TDNN(1024C) | × | - | - | - | - | - | - | - | - |
| | | ✓ | - | - | - | - | - | - | - | - |

## 4. RESULTS

### 4.1. Base Result

As is shown in table.2, we can see a significant improvement after introduce the VoxBlink-clean.

### 4.2. Ablation Result

#### *4.2.1. Dataset for training*

Use only VoxBlink for training.

#### *4.2.2. Strategy for training*

Compare the mix training and pretrain&finetune strategy.

#### *4.2.3. LMFT for training*

Large margin fine-tune (LMFT) strategy[29] has proved to be effective in ASV training. Thus we proceed to optimize the model by elevating the margin to 0.5 and expanding the duration to 6s till convergence. Table.3 shows our result based on SimAM-ResNet34, we observe that the LMFT benifits the performances on both Vox-O and VoxSRC23-dev protocols. Nevertheless, without incorporating VoxBlink for training, LMFT achieves a 12.7% improvement, while with the mixed fine-tuning using VoxBlink-clean, it shows only a 5.8% boost. We speculate that this might be because the average duration of VoxBlink is shorter than that of VoxCeleb2, meaning that longer segments actually contain limited information. This leads to a muted improvement when increasing the training duration.

**Table 3**. *Large-Margin fine-tune results based on SimAM-ResNet34 with/without the VoxBlink-clean data.*

| SimAM-ResNet34 | | | | | |
|---|---|---|---|---|---|
| VoxBlink-clean | LMFT | Vox-O | | VoxSRC23-dev | |
| | | EER | $mDCF_{0.05}$ | EER | $mDCF_{0.01}$ |
| × | × | 0.767 | 0.066 | 4.153 | 0.227 |
| × | ✓ | 0.670 | 0.586 | - | - |
| ✓ | × | 0.638 | 0.070 | 4.247 | 0.231 |
| ✓ | ✓ | 0.601 | 0.452 | - | - |

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